

Claire AI Mode Switching Architecture

****CLAIRE AI -- MODE-SWITCHING ARCHITECTURE Dynamic Governed Dual-Core Activation Secure Transition Framework****

1. Overview

Claire AI supports three operational modes:

1. ****Deterministic Mode**** Pure logic, air-gapped, patent-safe, reproducible.
2. ****Connected Intelligence Mode**** Live data ingestion, real-time scanning, dynamic intelligence.
3. ****Hybrid Mode**** Both engines active simultaneously, dual-core intelligence.

The Mode-Switching Architecture defines:

how Claire transitions between modes
how safety boundaries are enforced
how data flows are isolated or merged
how engines activate or deactivate
how governance controls mode selection

This ensures Claire can operate safely in:

sovereign environments
enterprise environments
regulated environments
dynamic strategic environments

2. Mode-Switching Principles

2.1 Isolation by Default

Each mode is isolated:

Deterministic Mode has zero external access.

Connected Mode has controlled external access.

Hybrid Mode has dual-pipeline access with governance.

Isolation ensures:

patent safety
data integrity

reproducibility

compliance

2.2 Explicit Activation

Claire never switches modes automatically. Mode changes require:

explicit user instruction

policy compliance

environment validation

governance approval (if configured)

2.3 One-Way Safety Boundaries

Certain transitions require safety checks:

Deterministic -> Connected Requires confirmation due to external data activation.

Connected -> Deterministic Requires flushing of live data buffers.

Hybrid -> Deterministic Requires full isolation reset.

2.4 Data Flow Governance

Each mode has strict data flow rules:

Deterministic: internal-only

Connected: external -> internal

Hybrid: external internal (governed)

Governance ensures:

no contamination

no leakage

no cross-mode drift

3. Mode-Switching Flow

3.1 Allowed Transitions

Claire supports the following transitions:

Deterministic -> Connected

Connected -> Deterministic

Deterministic -> Hybrid

Connected -> Hybrid

Hybrid -> Deterministic

Hybrid -> Connected

All transitions are explicit and governed.

4. Mode-Switching Architecture Diagram (Unicode)

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CLAIRE AI -- MODE-SWITCHING ARCHITECTURE Deterministic Mode Air-Gapped  
Patent-Safe Reproducible (Isolation Reset Required) Connected Mode Live  
Data Real-Time Signals Dynamic Scoring (Dual-Pipeline Activation) Hybrid  
Mode Dual-Core Engine Cross-Validation Max Capability (Governed  
Deactivation Paths) Deterministic Mode
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5. Transition Requirements

5.1 Deterministic -> Connected

Requires:

user confirmation

activation of external data pipelines

compliance check

environment validation

5.2 Connected -> Deterministic

Requires:

flushing live buffers

disabling external feeds

resetting semantic overlays

returning to internal-only logic

5.3 Deterministic -> Hybrid

Requires:

enabling Connected pipelines

enabling dual-core fusion layer

enabling hybrid scoring models

5.4 Connected -> Hybrid

Requires:

- enabling deterministic engine
- enabling cross-validation layer
- enabling hybrid semantic fusion

5.5 Hybrid -> Deterministic

Requires:

- disabling external feeds
- flushing live data
- isolating deterministic core
- resetting hybrid overlays

5.6 Hybrid -> Connected

Requires:

- disabling deterministic core
- maintaining live pipelines
- maintaining dynamic scoring

6. Governance Controls

6.1 Policy-Based Mode Restrictions

Organizations may restrict:

- Hybrid Mode
- Connected Mode
- Deterministic Mode only environments

6.2 Audit Logging

Every mode switch is logged:

- timestamp
- user
- environment
- reason

compliance state

6.3 Environment-Based Mode Enforcement

Examples:

Classified environment: Deterministic only

Enterprise R&D;: Hybrid

Market intelligence: Connected

7. Summary

The Mode-Switching Architecture ensures Claire can:

operate safely

transition cleanly

maintain data integrity

enforce compliance

support multiple environments

activate dual-core intelligence when needed

Hybrid Mode becomes possible only because the switching architecture is:

governed

explicit

isolated

validated

This architecture is essential for enterprise, federal, and acquisition-grade deployments.